AGUS SOBARNA

Mechatronics Engineer | Automation & Industry 4.0 Specialist

Bogor, West Java, Indonesia | +62 812-9173-1358 | sobarna.agus@gmail.com
linkedin.com/in/agus-sobarna-a4052658

PROFESSIONAL SUMMARY

Innovative **Mechatronics Engineer** with extensive professional experience in industrial automation, robotics, smart factories, and process optimization. Expertise in **SCADA systems, IoT, machine vision (OpenCV-Python)**, and utility management. Proven track record in reducing operational costs, implementing **Industry 4.0** solutions, and leading cross-functional engineering projects across palm oil industry, automotive, and chemical industries, managing comprehensive **CAPEX/OPEX** projects, and driving operational efficiency through innovative automation solutions. Recognized for bridging technical operational technology (OT) with high-level plant performance, safety, and strict regulatory standards.

TECHNICAL SKILLS & DOMAIN EXPERTISE

Automation & Robotics:
SCADA Systems, PLC Programming & Troubleshooting, Python (OpenCV and libraries), Robotic Integration, Special Purpose Machines (SPMs), Electro-Pneumatics, Motion Control.

Industry 4.0 & Smart Factory:
Smart Condition Monitoring, Overall Equipment Effectiveness (OEE) Tracking, Industrial IoT (IIoT) Deployment, Predictive Maintenance Protocols, Real-Time Production Analytics.

Utility Management:
Wastewater Treatment Plant (WWTP), Water Treatment Plant (WTP) & Steam Boilers, Sewage Treatment Plant (STP), Cleanroom HVAC & Chiller Systems, Wet Scrubbers, Dust Collectors, Fire Protection Systems.

Project Leadership:
CAPEX & OPEX Project Lifecycle, Lean Six Sigma (Green Belt), Failure Mode & Effects Analysis (FMEA), Root Cause Analysis, TPM, Cross-Functional Team Leadership, Technical Training.

Certifications & Standards:
BNSP Water Pollution Control, BNSP Air Pollution Control, SGS FSSC 22000, AIB Food Safety, EHEDG Hygienic Design, Kemenaker Fire Protection Class D, Kemenaker TKBT2 (Working at Heights).

PROFESSIONAL WORK EXPERIENCE

PT Firmenich Indonesia (DSM-Firmenich)

May 2022 – Present

Process Automation Engineer & Utility Supervisor | Bogor, West Java, Indonesia

Supervising plant-wide process automation, utility operations, and multi-disciplinary CAPEX projects for a multinational flavor and fragrance manufacturer, ensuring uncompromising compliance with international food hygiene (EHEDG, FSSC 22000) and environmental safety standards.

- **Critical Process Automation:** Engineered and commissioned automated transfer systems for high-viscosity PG/DPG materials, substantially reducing batch cycle times, eliminating manual handling strain, and completely eliminating human operational errors.
- **Turnkey CAPEX Modernization:** Directed cross-functional CAPEX execution covering chiller plants, cleanroom HVAC systems, wet scrubbers, dust collection units, fire sprinkler networks, and WWTP capacity expansion—completing all strategic facility targets on schedule.
- **Utility Performance & Energy Optimization:** Optimized facility utility operations (HVAC, WWTP, STP, and WTP steam systems) through automated utility monitoring and predictive maintenance, cutting plant energy consumption significantly.
- **Asset Reliability & Predictive Maintenance:** Developed and executed preventive and predictive maintenance programs across critical utility infrastructure (WWTP, STP, HVAC, fire hydrant water), substantially reducing equipment downtime and elevating asset operational life.
- **Environmental & Regulatory Compliance:** Ensured strict adherence to environmental regulations as certified custodian for BNSP Water and Air Pollution Control, maintaining zero non-conformances in environmental audits.

PT SMART Tbk (Sinarmas Agribusiness and Food)

2019 – 2022

Workshop & Development Supervisor | Indonesia

Led engineering workshop operations, research & development of custom estate automation machinery, and agricultural modernization initiatives across extensive palm oil plantations.

- **AI-Driven Agricultural Modernization:** Pioneered AI-powered palm oil automation by developing a custom OpenCV + Python computer vision algorithm to identify and geolocate palm oil trees from drone aerial imagery, integrating detection targets with an auto-fertilizer spreader to drastically reduce field labor dependency and enhance distribution accuracy.
- **Specialized Machinery Innovation:** Designed, built, and commissioned innovative specialized machinery tailored for rugged palm oil estate environments, slashing substantial recurring operational expenditures.
- **Workforce Capability Building:** Formulated and conducted technical training programs for field engineers and technicians in control systems, electro-pneumatics, and hardware-software system integration, elevating workforce autonomy.

PROFESSIONAL WORK EXPERIENCE (CONTINUED)

PT Astra Otoparts Tbk

2013 – 2019

Smart Factory Engineer | 2017 – 2019

Spearheaded digital transformation and smart manufacturing initiatives across high-volume automotive component manufacturing plants.

- **SCADA & OEE Telemetry:** Implemented centralized SCADA-based OEE monitoring across multiple assembly lines, markedly boosting equipment effectiveness and establishing real-time operational visibility for production supervisors.
- **Industrial IoT Implementation:** Deployed non-invasive edge IoT solutions for live production tracking and machine condition telemetry, substantially reducing unplanned production downtime and streamlining maintenance intervention.

Electrical & Robotic Design Engineer | 2013 – 2017

Engineered electrical systems, control architectures, and robotic automation for specialized manufacturing equipment supplied to automotive clients.

- **Special-Purpose Machine (SPM) Engineering:** Designed custom SPMs and robotic automation cells for automotive clients, significantly improving production assembly speeds while adhering to strict automotive quality and cycle standards.
- **Failure Mode & Effects Analysis (FMEA):** Conducted comprehensive FMEA across diverse machine lines, mitigating potential design hazards and consistently ensuring on-time project delivery with zero warranty defects.

KEY PROJECT HIGHLIGHTS

1. AI-Powered Fertilizer Spreader

PT SMART Tbk

Challenge & Solution: Replaced physically grueling manual fertilizer distribution across large-scale plantations by engineering an intelligent computer vision algorithm (OpenCV + Python) that analyzes drone aerial imagery to detect and map palm oil trees.

Outcome: Successfully linked tree coordinates to automated fertilizer spreader machinery, achieving high detection accuracy, substantial labor cost reductions, and precise nutrient delivery.

2. SCADA Automation for Chemical Transfer

PT Firmenich Indonesia

Challenge & Solution: Modernized manual chemical handling of Propylene Glycol / Dipropylene Glycol (PG/DPG) by engineering an automated closed-loop tank transfer system with mass flow control, actuated valves, and real-time SCADA interlocks.

Outcome: Completely eliminated manual handling errors, drastically reduced batch cycle times, and ensured full compliance with FSSC 22000 and EHEDG hygienic safety protocols.

3. Smart Factory IoT Deployment

PT Astra Otoparts Tbk

Challenge & Solution: Eliminated blind spots in production tracking across automotive component lines by instrumenting machinery with edge IoT telemetry gateways linked to centralized SCADA systems.

Outcome: Integrated live OEE and downtime visualization dashboards, accelerating maintenance troubleshooting and substantially improving overall production uptime.

KEY ACHIEVEMENTS

- **On-Time CAPEX Execution:** Multi-scale capital expenditure modernization projects successfully delivered strictly on schedule through disciplined engineering management and contractor governance.
- **Estate Labor Modernization:** Achieved substantial labor reduction and operational efficiency in palm oil estates via proprietary computer vision and machinery automation.
- **Sustainable Energy Optimization:** Achieved significant facility-wide energy savings through systematic utility optimization, predictive maintenance, and steam system balancing.

EDUCATION

Diploma 3 (A.Md.) in Automation & Mechatronics Engineering

Graduated: 2013

Politeknik Manufaktur Negeri Bandung (PMS - ITB)

Comprehensive vocational engineering program centered on industrial automation, electro-pneumatics, precision machine design, PLC programming, microcontrollers, and manufacturing technology.

LICENSES & PROFESSIONAL CERTIFICATIONS

Water Pollution Control (PPPA)	BNSP	Air Pollution Control (PPPU)	BNSP
FSSC 22000 (Food Safety System)	SGS	Food Safety Standards	AIB Intl
Hygienic Design & Engineering	EHEDG	Fire Protection Class D	Kemenaker RI
TKBT2 (Working at Heights)	Kemenaker RI	Lean Six Sigma (Green Belt)	Certified